Differential transcript usage analaysis for dynamic biological processes

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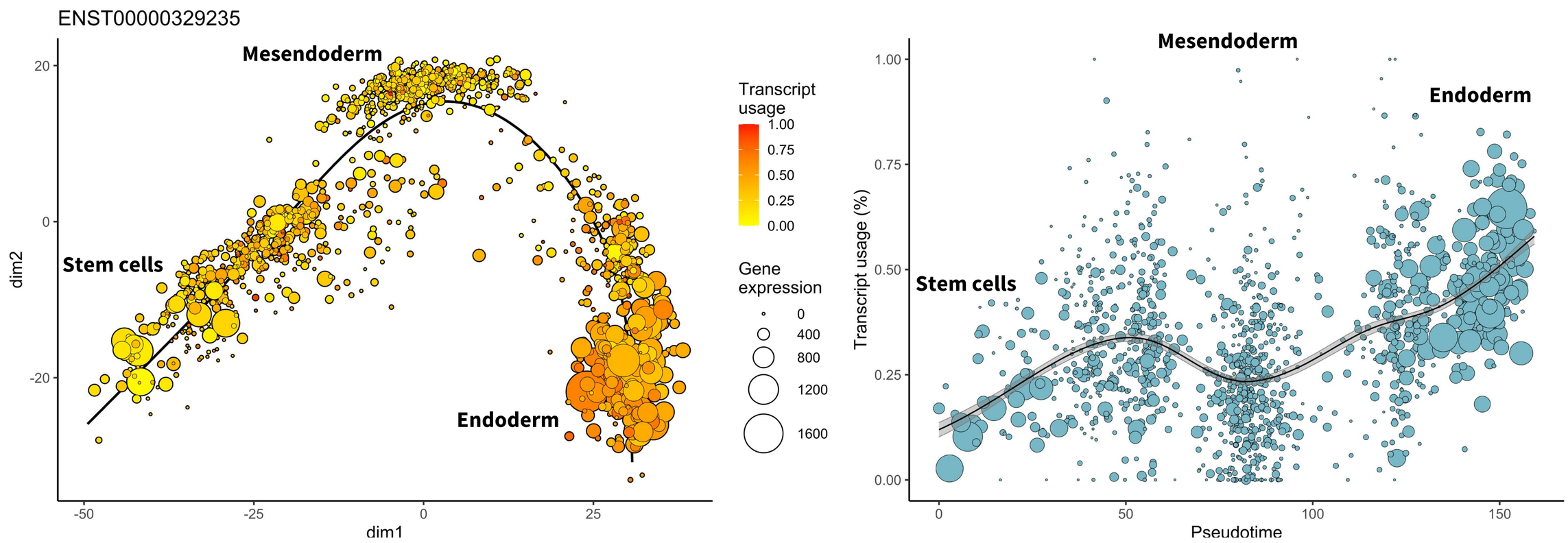
Highlights

- tradeSeq tests differential gene/transcript expression along trajectories
- We now extend tradeSeq for testing differential transcript usage along trajectories
- Powerful visualizations to inspect differential transcript usage signal

Background

- Trajectory inference has been instrumental for modelling dynamic changes in biological systems
- Downstream, a key challenge is the identification of biological markers.
- Our Bioconductor package **tradeSeq** performs differential expression tests for genes or transcripts, both within and between lineages
- One **drawback** of tradeSeq is that it cannot directly test for differential transcript usage (DTU, the change in relative usage of transcripts within the same gene)
- Several Bioconductor packages like DEXSeq, DRIMSeq and our package satuRn implement DTU tests, but are limited to comparisons between discrete groups of cells.
- **We here extend tradeSeq for testing DTU along trajectories.**



Differential transcript usage during human endoderm development. **Left panel:** Cells at different stages of differentiation in UMAP space. The usage of transcript ENST00000329235 gradually increases during differentiation of induced pluripotent stem cells to endoderm cells, as indicated by the color scheme. Dot sizes correspond to the expression level of the corresponding gene of the transcript. **Right panel:** Alternative representation. Transcript usage increases over pseudotime, as displayed by the black smoother inferred by the quasi-binomial GAM.

Statistical model

- Take Y_{gti} the expression of transcript t of gene g in cell i .
  The total expression of gene g in sample i can then be expressed as:

$$Y_{g.i} = \sum_{t=1}^{T_g} Y_{gti}$$

- Denote the usage of transcript t of gene g in sample i as:

$$U_{gti} = \frac{Y_{gti}}{Y_{g.i}}$$

- Describe the quasi-binomial generalised additive model (GAM) as:

$$\begin{cases} E[U_{gti}|\boldsymbol{X_i}, Y_{g.i}] = \pi_{gti} \\ \log\left(\frac{\pi_{gti}}{1-\pi_{gti}}\right) = \eta_{gti} \\ \eta_{gti} = \sum_{l=1}^{L} s_{gtl}(T_{li}) Z_{li} + \boldsymbol{U_i \alpha_{gt}} \end{cases}$$

Smooth function of pseudo-time for each lineage — Assigns cells to lineages — Covariates (fixed effects)

- With variance

$$Var[U_{gti}|\boldsymbol{X}_i, Y_{g.i}] = \frac{\pi_{gti}(1-\pi_{gti})}{Y_{g.i}} \phi_{gt}$$

Hypothesis testing

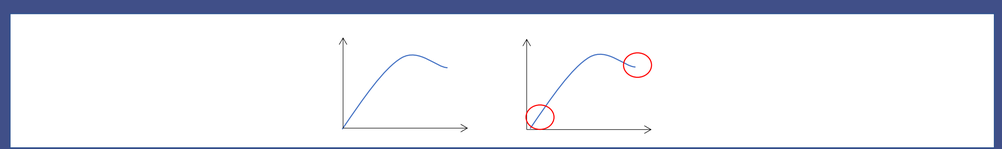
- The smoothing spline s_{gl}, for a given transcript t of a gene g and lineage l can be represented as a linear combination of K cubic basis functions

$$s_{gl}(t) = \sum_{k=1}^{K} b_k(t) \beta_{gtlk}$$

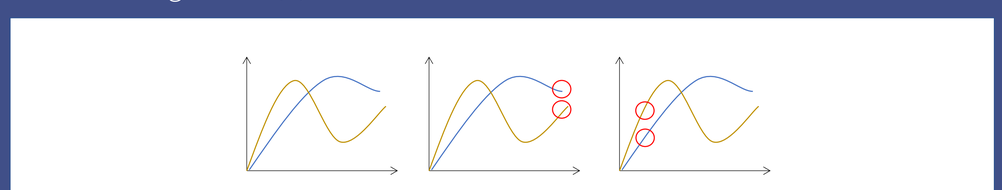
- Testing for differential usage then involves testing linear combinations of the parameters β_{gtlk} using Wald tests

$$W_{gt} = \hat{\beta}_{gt}^T C (C^T \hat{\Sigma}_{\hat{\beta}_{gt}} C)^{-1} C^T \hat{\beta}_{gt}$$

- This flexible framework allows for testing five* distinct hypotheses
  - Within lineage tests (2)



  - Between lineage tests (3)



References
tradeSeq: Van den Berge et al., Nature Communications 11, 1201 (2020)
satuRn: Gilis et al., F1000 Research, 10:374 (2022)
Case study data: Cuomo et al., Nature Communications 11, 810 (2020)